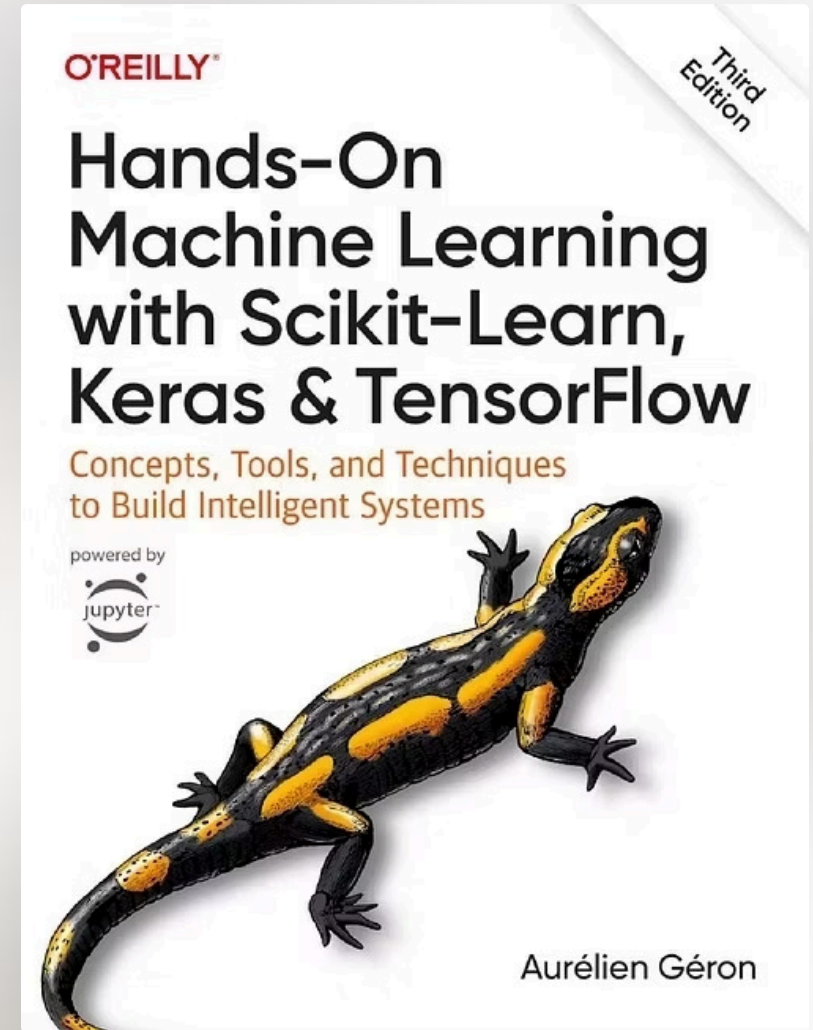


Machine Learning

Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

RP by Rodrigo Martins Pagliares

Last update: 08/08/2025



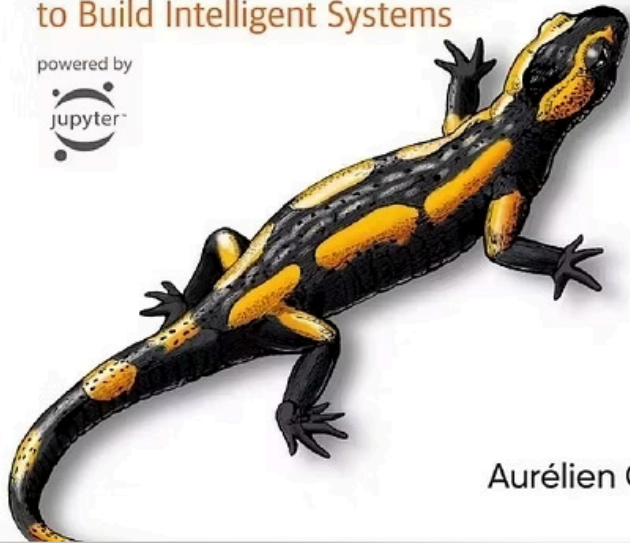
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Third
Edition

Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow

Concepts, Tools, and Techniques
to Build Intelligent Systems

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Aurélien Géron

Machine Learning

Chapter 1 – The Machine Learning Landscape

Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. Types of Machine Learning Systems
5. **Main Challenges of Machine Learning**
6. Testing and Validating





Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. Types of Machine Learning Systems
 - Classification criteria #1: Supervision during training
 - Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning
 - Classification criteria #2: Approach to Learning Method
 - Batch and Online (Incremental) Learning
 - Classification criteria #3: Learning Method
 - Instance-Based and Model-Based Learning
5. Main Challenges of Machine Learning
 - Insufficient Quantity of Training Data
 - Non-representative Training Data
 - Poor-Quality Data
 - Irrelevant Features
 - Overfitting the Training Data
 - Underfitting the Training Data
 - **Stepping Back: Reviewing the Chapter So Far**
6. Testing and Validating
 - Hyperparameter Tuning and Model Selection
 - Data Mismatch

Main Challenges of Machine Learning

Insufficient Quantity of Training Data, Non-representative Training Data, Poor-Quality Data, Irrelevant Features, Overfitting the Training Data, Underfitting the Training Data, **Stepping Back: Reviewing the Chapter So Far**

Stepping Back: Overview of ML

Big Picture:

ML is about getting machines to **improve at tasks** by **learning from data** instead of **explicitly coding rules**.





Types of Machine Learning Systems

1

Supervised vs. Unsupervised:

Supervised Learning:

Trains models with labeled data.

Unsupervised Learning:

Finds patterns in unlabeled data.

2

Batch vs. Online:

Batch Learning: Trains on all data at once.

Online Learning: Trains incrementally as new data arrives.

3

Instance-Based vs. Model-Based:

Instance-Based Learning:

Learns examples by heart and generalizes using similarity measures.

Model-Based Learning:

Trains a model by adjusting parameters to fit the data.

The Machine Learning Process

1. Gather Data

Collect the training set for the learning algorithm.

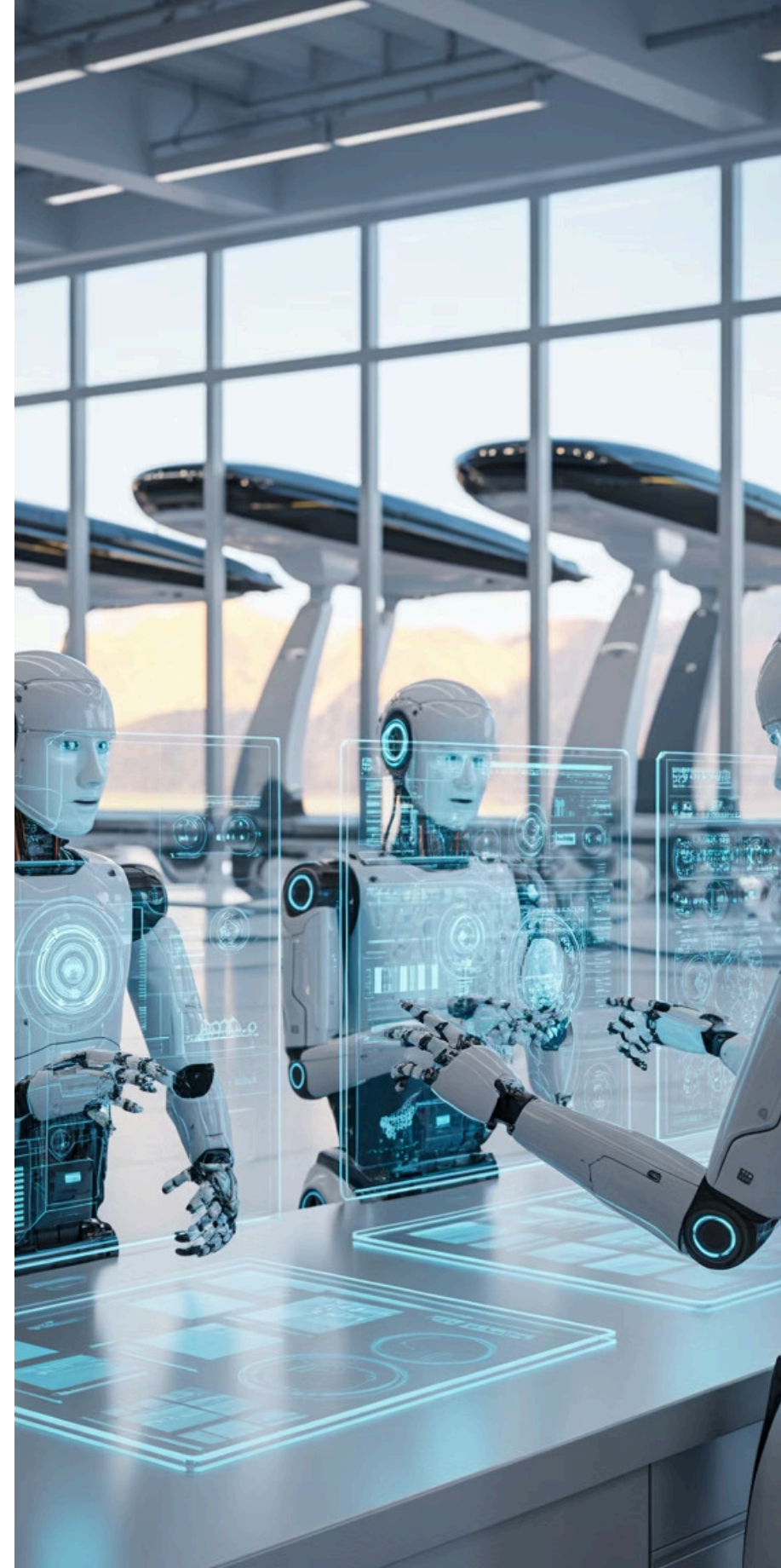
2. Train the Model

Model-Based Algorithms: Adjust parameters to fit the model to the training set.

Instance-Based Algorithms: Learn examples by heart and generalize based on similarity.

3. Generalization:

The model should ideally make **good predictions** on **new cases** beyond the training set.



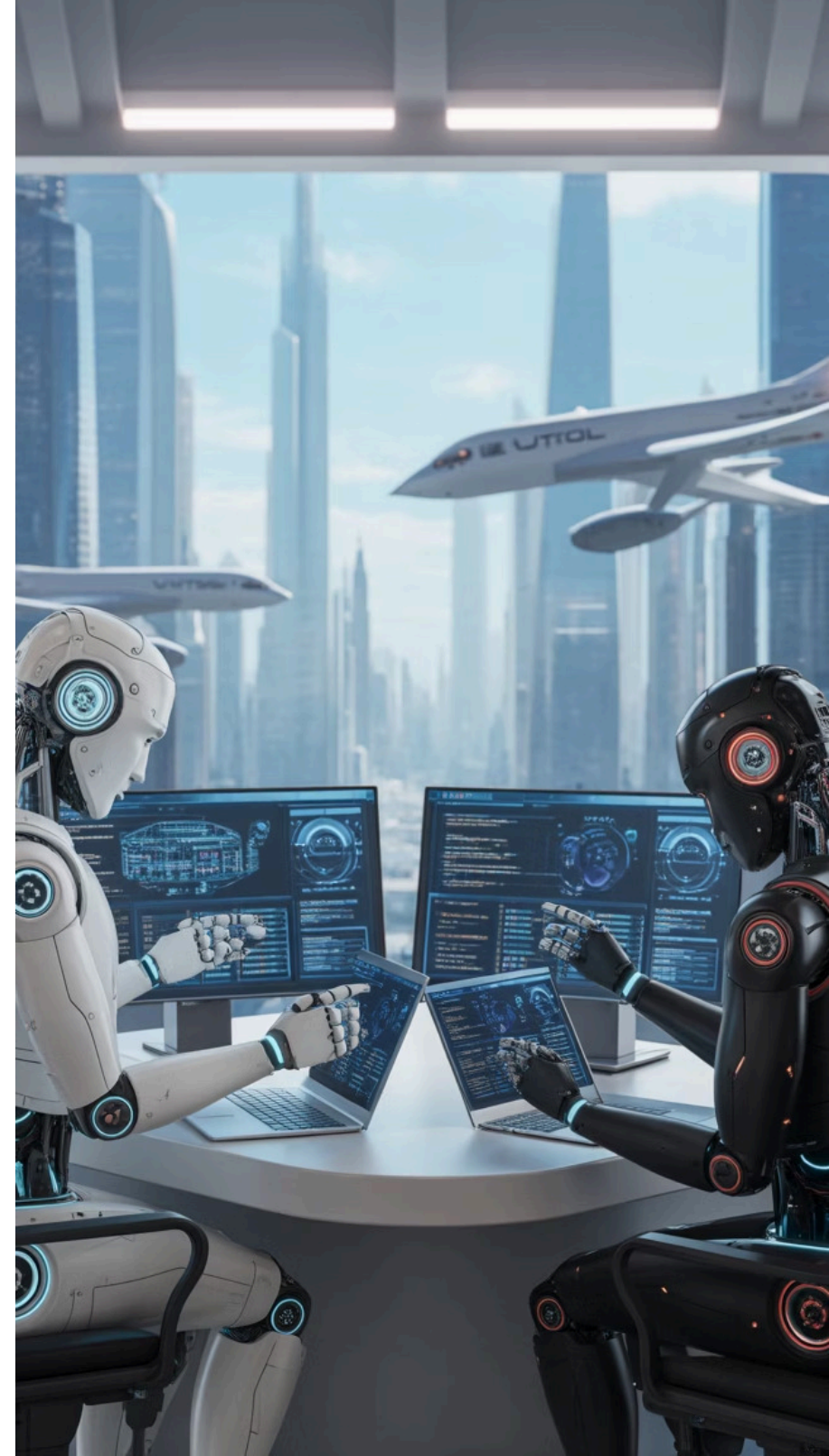
Common Challenges in Machine Learning

Data Issues:

- **Small datasets** may lead to poor model performance.
- **Non-representative data** or **noisy data** can lead to inaccurate predictions.
- **Irrelevant features** (garbage in, garbage out) can reduce model effectiveness.

Model Complexity:

- **Underfitting:** A model that is too simple will fail to capture data patterns.
- **Overfitting:** A model that is too complex will memorize the training data and fail to generalize.



Model Evaluation and Fine-Tuning

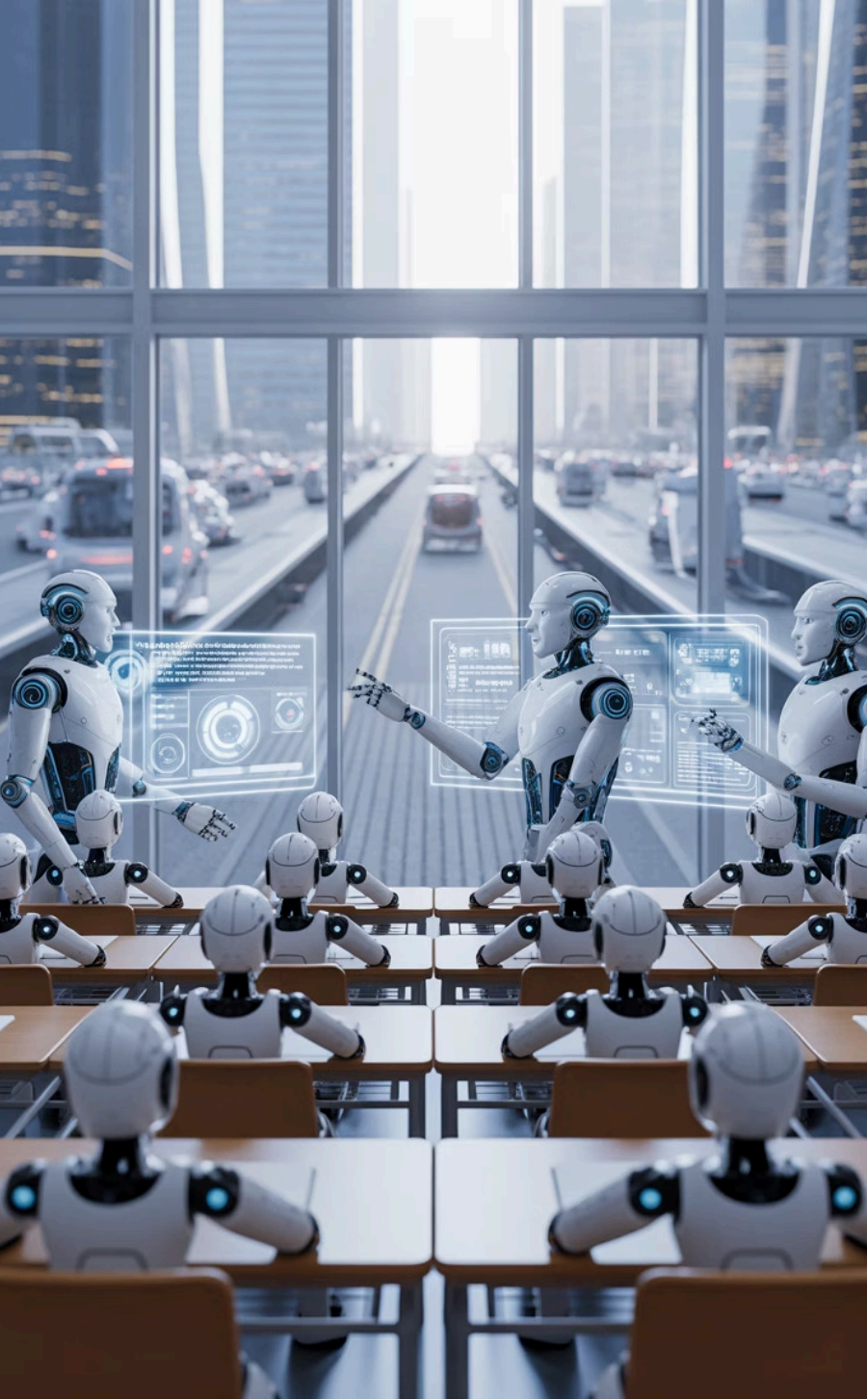
Evaluation:

Once the model is trained, it must be **evaluated** to ensure it **generalizes** well to **new, unseen data**.

Fine-Tuning:

Fine-tune the model by **adjusting hyperparameters** and **re-evaluating** to **improve performance**.





Key Takeaways

ML Process:

Gather data, train models, and ensure generalization to new cases.

Challenges:

Pay attention to data quality and model complexity to avoid **underfitting** or **overfitting**.

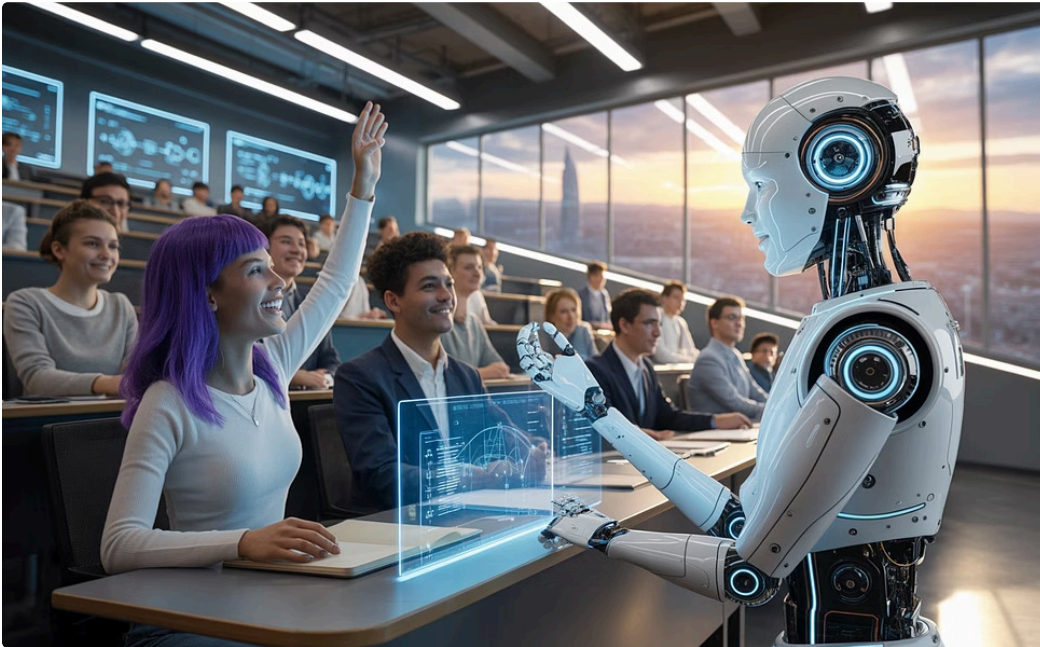
Evaluation and Fine-Tuning:

Always **evaluate** the model's performance on **unseen data** and **fine-tune** it as necessary to improve accuracy.

Next: Testing & Validating

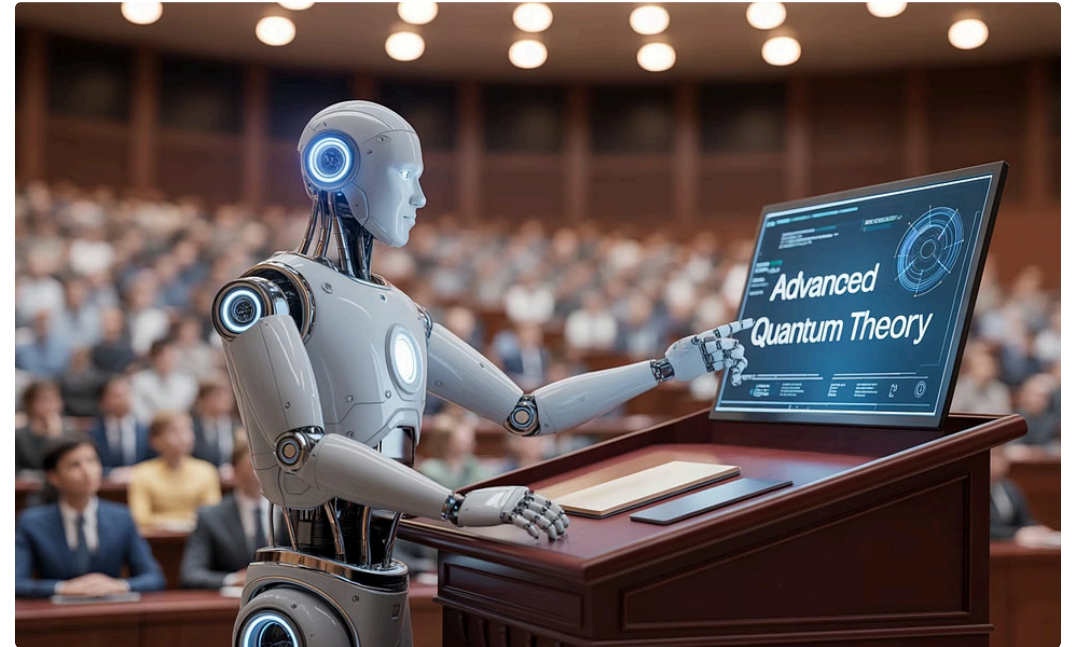


Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts